

IT Infrastructure Library (ITIL) and Configuration Management Databases (CMDBs)

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What did we learn last hour? – Repetition

Explain the following terms:

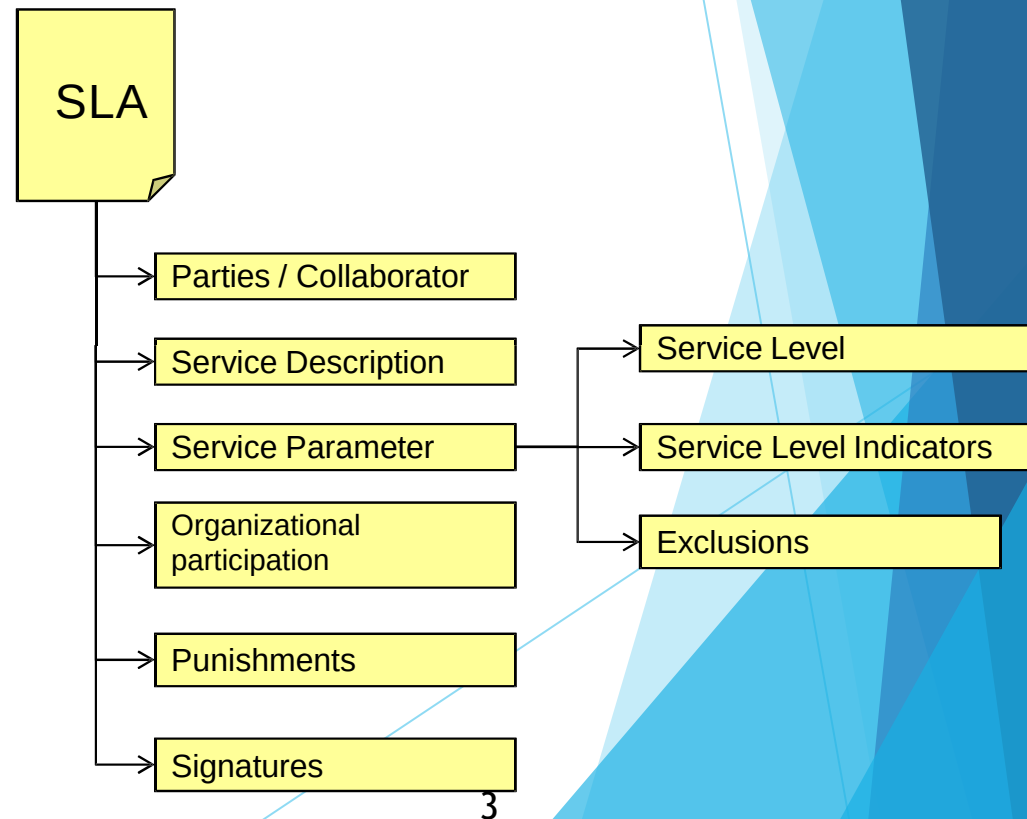
- Define network management activities and precautions
- Basic idea "Network as factory"
- Value chain
- Business process

Objective and structure of today's lecture

- Overall picture of IT service management based on ITIL
- Overview of essential ITIL processes
- Introduction to Configuration Management Database (CMDB)

Service Level Agreement

- User as a customer of the IT organization. The customer wants ...
 - to be understood and taken seriously
 - an optimal satisfaction of his needs
 - transparency regarding the services provided
- => Service Level Agreement (SLA)



What's IT Infrastructure Library (ITIL)?

- ▶ ITIL is a framework of best practices approaches (in terms of processes, guidelines, checklists etc.) intended to facilitate the delivery of high quality information technology services.
 - ▶ Naming of processes
 - ▶ Structuring and grouping of processes
 - ▶ Generic process flows
- ▶ ITIL is not
 - ▶ ready to use
 - ▶ directly implementable
- ▶ ITIL demands
 - ▶ adaptation to the respective company
 - ▶ extensive preparatory work
- ▶ Best Practices for IT Service Management Worldwide Industry standard
- ▶ Management Philosophy

Suppliers

Service Provider (IT Services)

Service Consumer

Software

Hardware

Personal

Building

**External
Service
Provider**

Transfer

Strategical

Business Perspective

Business Perspective Processes

Tactical

Appl. Mgmt.

Requirements

Design

Build

Deploy

**Operate &
Optimize**

Service Delivery

**Capacity-
Mgmt.**

**Availability-
Mgmt.**

**Continuity-
Mgmt.**

**Financial-
Mgmt.**

**Service-
Level-
Mgmt.**

Customers

Operational

Infrastr. Mgmt.

Design & Plan

Deployment

Operations

Techn. Support

Service Support

**Release-
Mgmt.**

**Change-
Mgmt.**

**Problem-
Mgmt.**

**Incident-
Mgmt.**
Service
Desk

**Configuration-
Mgmt.**

Users

Brief description of the processes

Processes of Service Delivery

- **Service Level Management:**
Set up and monitor service level agreements (SLAs)
- **Availability Management:**
Ensure and optimize service availability
- **Continuity Management:**
Restore key services in the event of a disaster
- **Capacity Management:**
Plan and monitor the resource capacity used for service delivery
- **Financial Management:**
Create transparency in terms of IT costs, budget planning

According to: Dr. Igor Radisic et al., Prof. Dr. Heinz-Gerd Hegering, TU München, Einführung in IT Service Management Prozesse

Brief description of the processes

Processes of Service Support

- **Service Desk:**
Single-point-of-contact to the user
- **Incident Management:**
As soon as possible restore the service in case of failure
- **Problem Management:**
Root cause analysis and problem solving for recurring failures
- **Change Management:**
Controlling changes with minimization of related disruptions
- **Release Management:**
Release, control and implementation of (software) release changes
- **Configuration Management:**
Provision and maintenance of a logical model of the IT Infrastructure and IT Services (-> CMDB)

Who uses ITIL?

- It has over 100.000 certified professionals and consultants.
- Europe, USA, Australia and Canada
- Typical organizations in USA that are implementing ITIL:

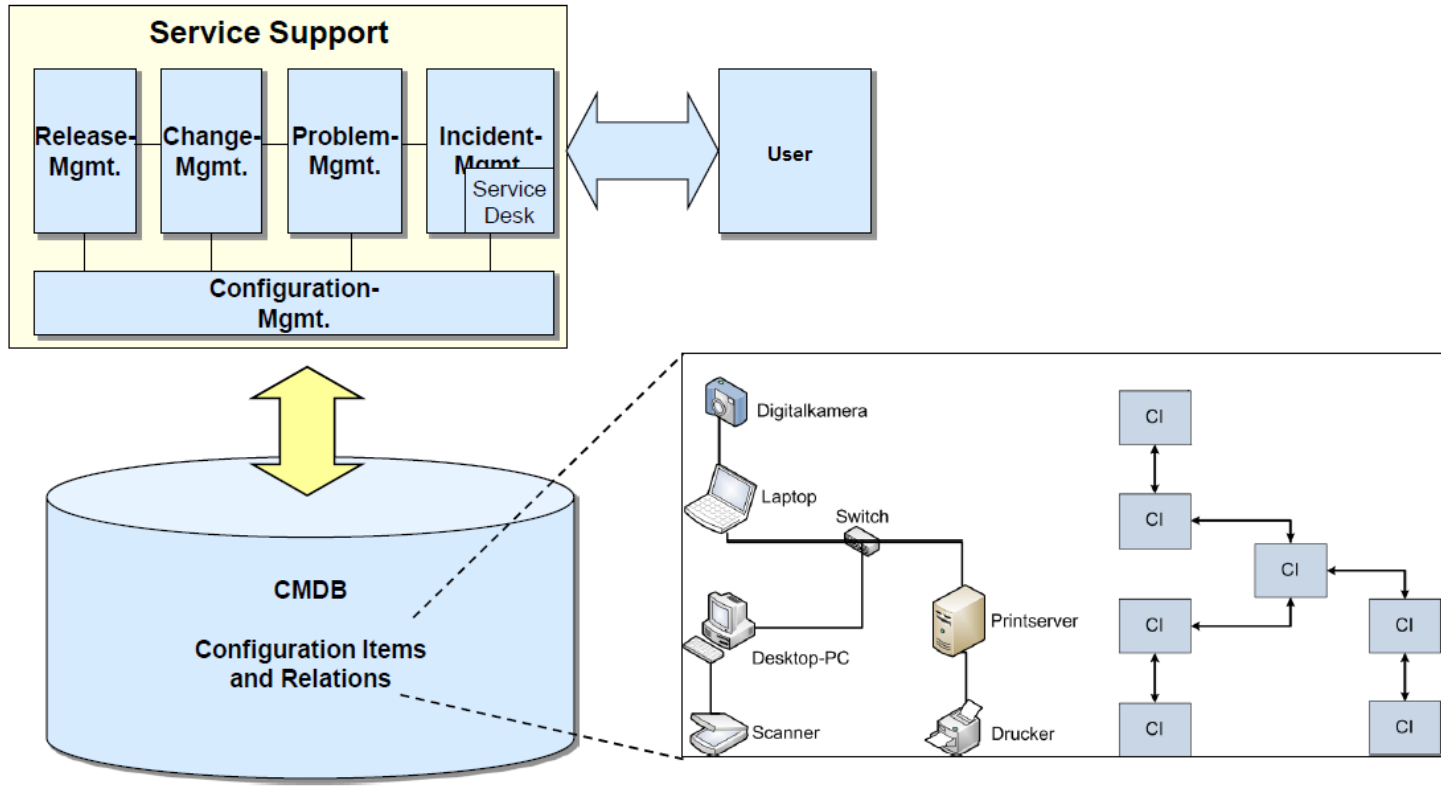
- Microsoft—by providing Microsoft Operations Framework (MOF)
- HP—by providing IT Service Management Reference Model
- IBM—by providing IT Process Model
- US Army
- State of California
- State of North Carolina
- Blue Cross – Blue Shield of Florida
- Blue Cross – Blue Shield of Texas
- LG&E Energy LLC
- United Health Group in Minneapolis

List taken directly from (Ball, 2005)

http://images.globalknowledge.com/wwwimages/whitepaperpdf/WP_ITIL.pdf

The IBM logo, consisting of the letters "IBM" in a blue, horizontally-striped font.The Microsoft logo, featuring the word "Microsoft" in a white, sans-serif font on a black background.

Configuration Management Database (CMDB)



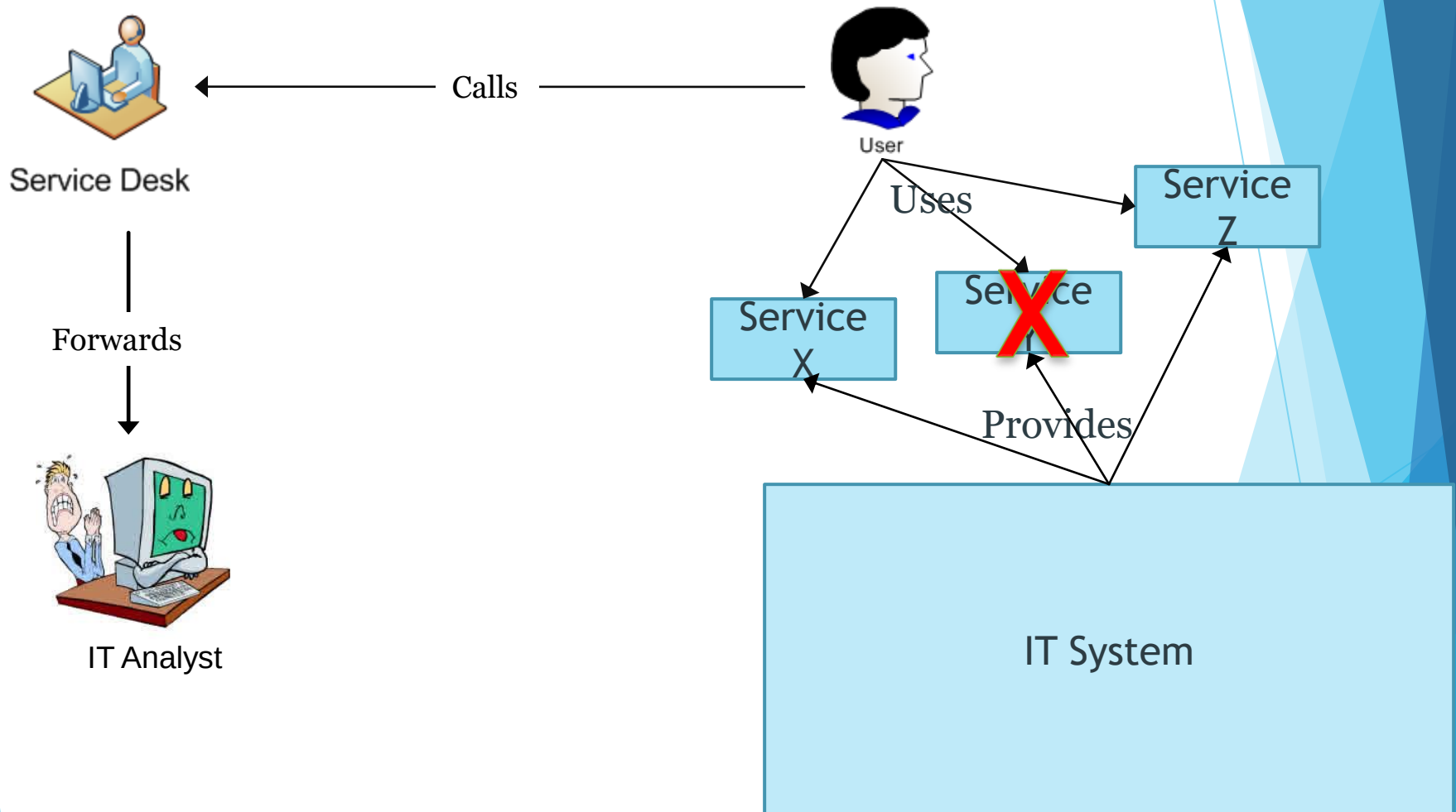
What is a Configuration Item (CI)?

- Component that is necessary to provide an IT:
 - Organizational unit
 - PC
 - Information system
 - Licenses
 - Cluster
 - Server
 - LAN
- Also, an entity involved in the provision of an IT service:
 - IT service itself
 - Users and customers
 - Business processes
 - Problems
 - Events / errors

Attributes and Relationships of the Configuration Items (CIs)

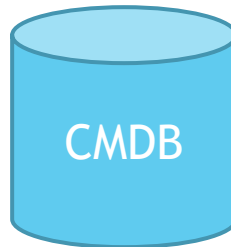
- Attributes of Configuration Items (CIs)
 - unique identifier
 - CI type
 - serial number
 - model number
 - warranty period
 - installation date
 - Administrator / CEO
 - Manufacturer
 - Status (active, out of service, in planning, in the test, in the warehouse, in repair, ...)
- Relationships between Configuration Items (CIs)
 - A is part of B (for example, a processor as part of a PC)
 - A is connected to B (for example, a printer is connected to a PC)
 - A triggers B (for example, a machine starts a service on a server)
 - A uses B (e.g., software instance uses a PC)
 - A depends on B (for example, server is dependent on UPS)

Configuration Management Databases (CMDB) - The big picture



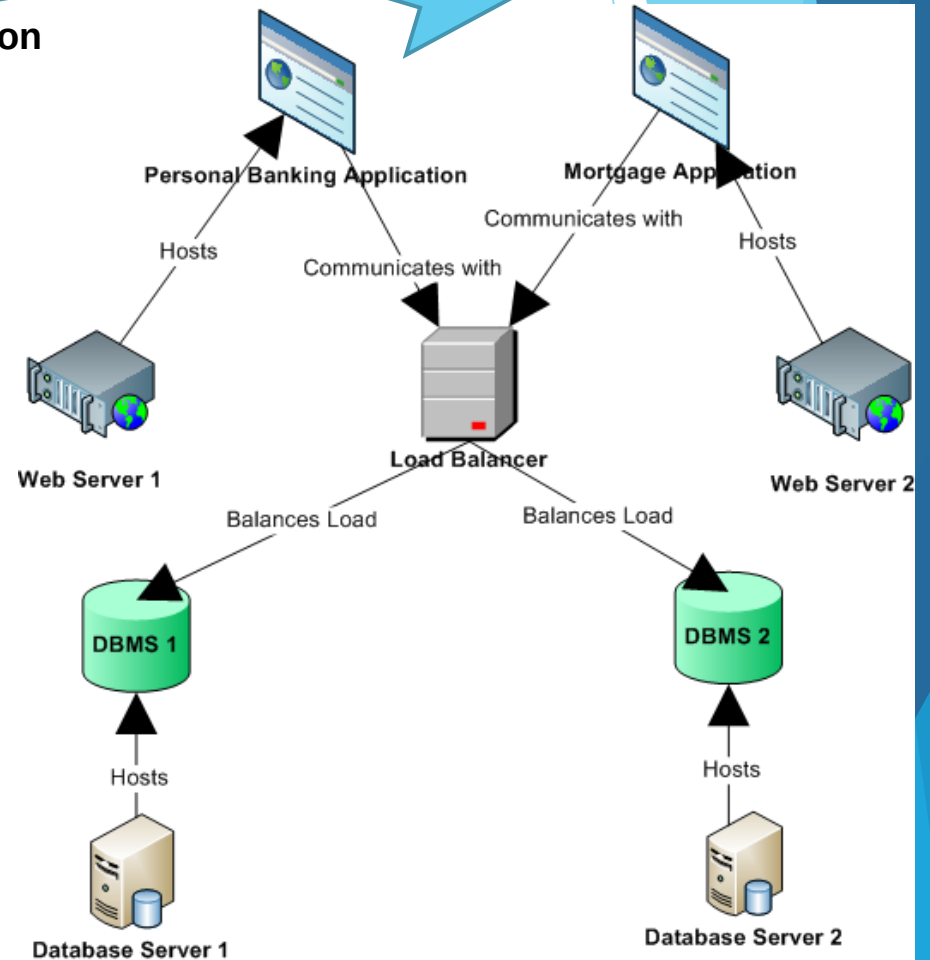


Has

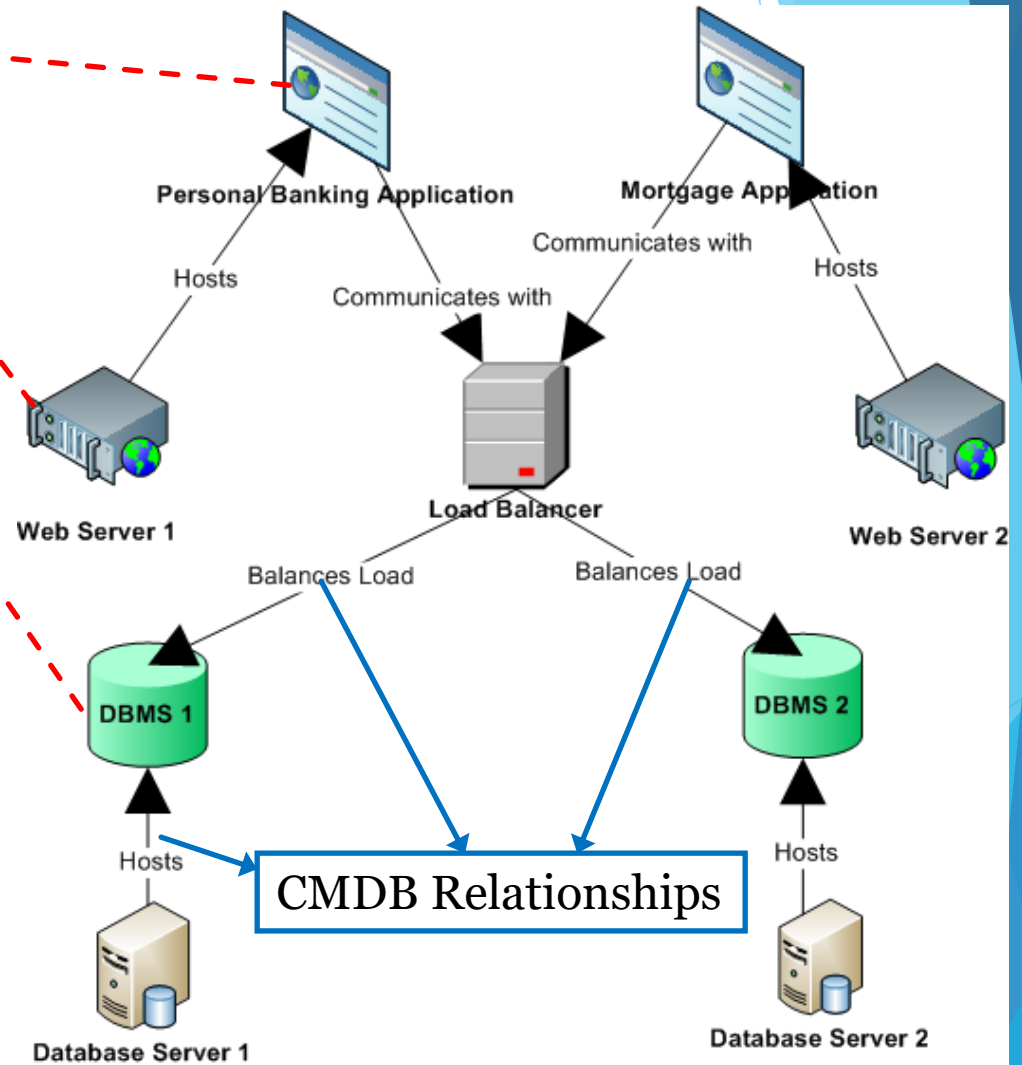


Information

Provides More Details



Configuration Items (CIs)



Knowledge contained in a CMDB

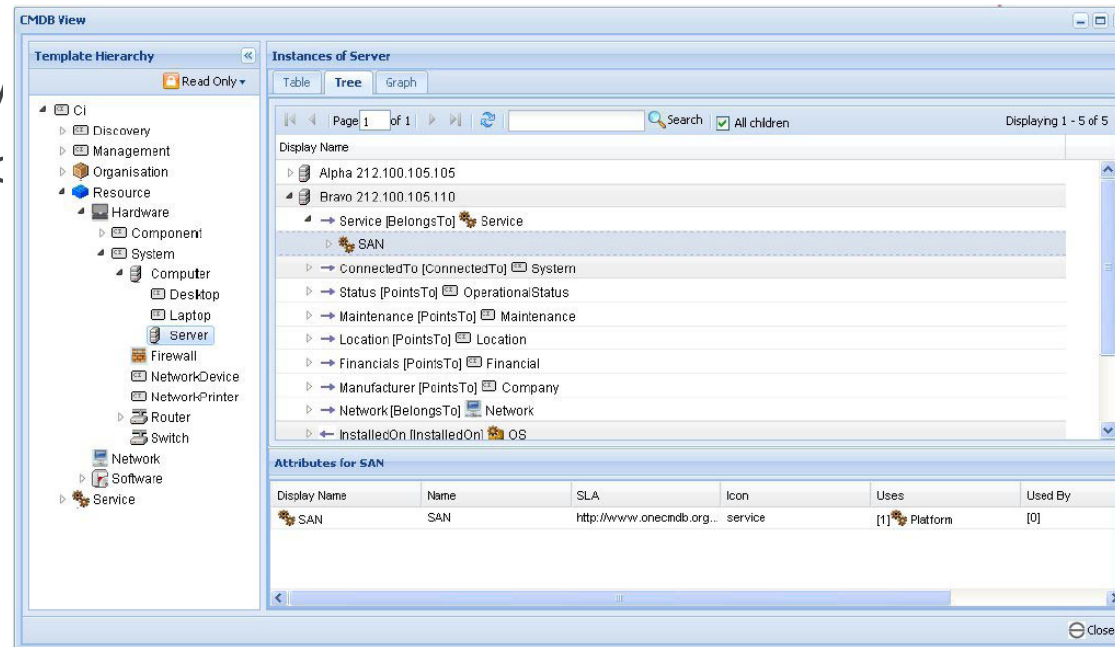
- ▶ Hardware, networks, locations, etc.
- ▶ Software, SLA, versioning information.
- ▶ Reporting structures, contacts, organization.
- ▶ Dependencies between components.
- ▶ Incident and Problem reports.
- ▶ History of changes made, by who, when, why, etc.
- ▶ Gateway to logs and diagnostics.

Requirements for a CMDB

- ▶ Visualization of the CMDB database
- ▶ Plausibility
- ▶ Right concept and user administration
- ▶ History of all guided components
- ▶ Query functions (selective and relational queries)
- ▶ Different views for different users
- ▶ Automatic collection of data through auto discovery
- ▶ Catalog of predefined data type templates

OneCMDB Overview

- ▶ Create your own CMDB data model, without writing a single line of code
- ▶ Populate the CMDB, through auto-discovery of your network
- ▶ Import data from other data sources (MDRs, Management Data Resources)
- ▶ Validate imported data and decide if it shall be inserted in the CMDB or not
- ▶ Visualize your data with dynamic views
- ▶ Export data to other applications



OneCMDB Key Features

- ▶ Flexible Data Model
- ▶ Object-oriented database
- ▶ Automatic change log
- ▶ Multi-user system
- ▶ Role Based Access Control
- ▶ Open APIs

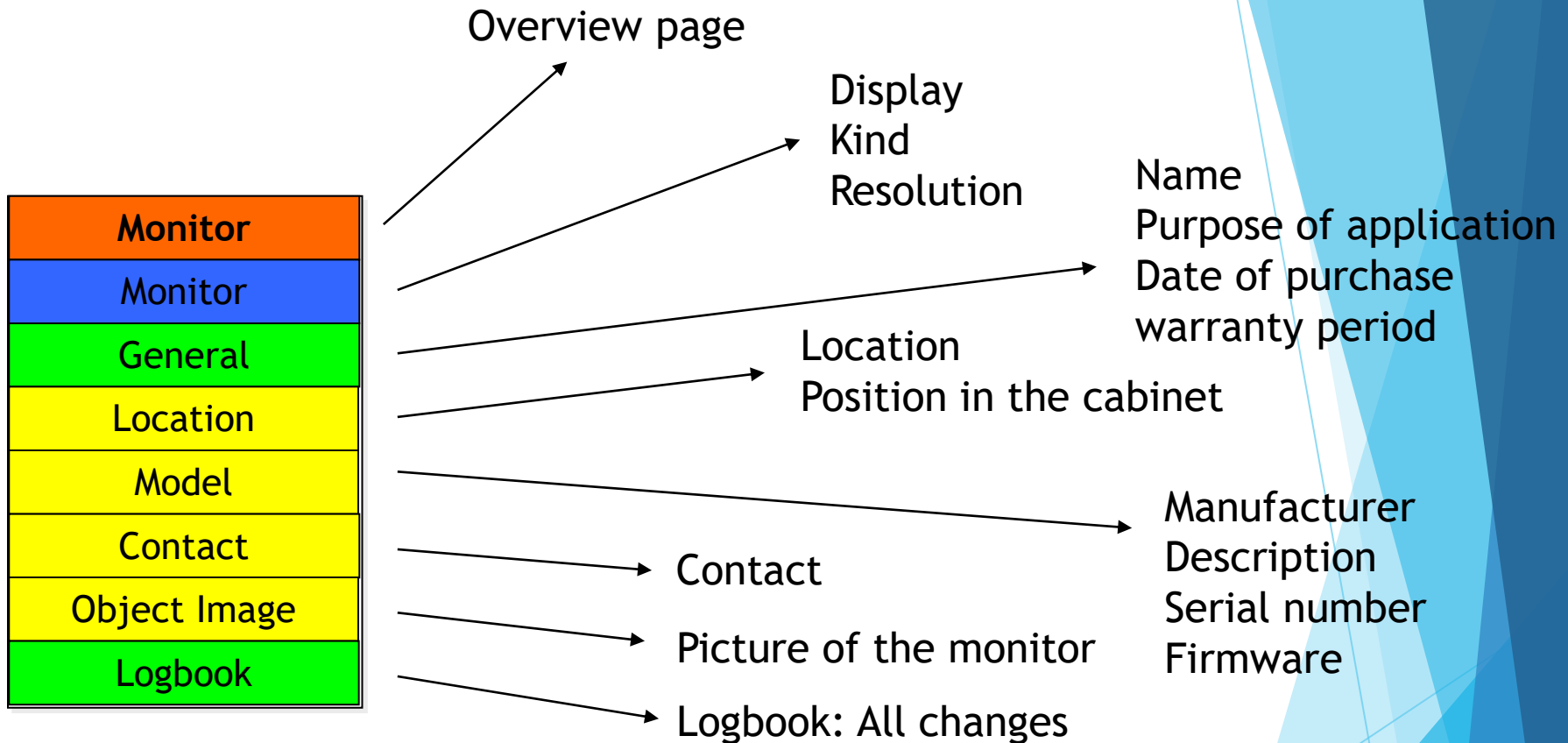
Supported/interfaced products

Hardware platforms:	Intel, Sparc
Operating systems:	Linux, Windows
Auto discovery systems:	NMAP
Network & systems management systems:	Nagios, Netcool
Relational databases:	HSQL, MySQL, MS-SQL, Oracle

i-doit CMDB

- ▶ The CMDB of i-doit consists of object types (templates)
- ▶ Object types consist of categories
- ▶ Categories have attributes or references
- ▶ Types of categories
 - ▶ General categories
 - ▶ Special categories
- ▶ Own object types based only on existing categories

Create object type (template)



- Special category
- Mandatory category
- General category

Existing Categories

General Categories		Special Categories	
Overview	Model	Building	WAN Lines
General	Form Factor	Room	Emergency Plan
Logbook	Dependencies	Cabinet	Maintenance
Application/Service	Maintenance Contract	Switch	FC-Switch
CPU	Files	Router	Stream object
Storage	Data Backup	WLAN	Network
Interfaces	Infocenter	Client	
Manual	Licences	Printer	
Access	Consumer	SAN	
Object Image		Monitor	
Contact		Air Conditioner	
Location		Mobile Phone	

 Mandatory Category

Object view vs. Location view

- ▶ Object view: ordered by object type
- ▶ Location view: sorted by location
 - ▶ Creation of container objects necessary
 - ▶ Example: Location view for a switch
 - ▶ Building: Computer Science
 - ▶ Room: C055
 - ▶ Cabinet: rack1
 - ▶ Switch: Netgear GSM7324

I-doit: Links

- ▶ Links of objects can be realized in many ways:
 - ▶ References (e.g., application and licenses)
 - ▶ Data Networks (interface, port)
 - ▶ Electricity networks (electricity object, consumers)
 - ▶ Storage networks (controller, drive)
 - ▶ Interfaces
 - ▶ Dependencies

Thank you for your attention!